

Biougra Pond: Human-Qubit-Machine Protocol
Version 1.0 - October 2026

Abstract

Biougra Pond is a minimal quantum-anchored timestamping protocol that binds a geographic coordinate, a Bitcoin block height, and a verifiable Bell-state measurement from an IBM Quantum QPU into a hash chain.

1. Origin Coordinate

- Location: 30.12613,-9.37437 (Biougra, Morocco)
- Sun elevation: 36.98 at timestamp
- Time: 2026-10-02T16:12:51.736920Z
- BTC Height: 969613

2. Quantum Proof

Circuit: biougra.qasm - Bell state preparation $H(0)$; $CX(0,1)$

Execution: real_qpu.py on IBM Quantum backend

Shots: 1024

Result: {'00': 492, '11': 532}

Verification: Correlated counts confirm entanglement, non-classical distribution.

3. Chain Structure

File: chain.json

Fields: pond, gps, gpsStr, btcHeight, prev hash, payload hash

Example hash: ab5446ede0c96a930f31307b92dbf386ea371f13774bb187a116412cb303918f

Prev: 3ae3d838f894c251e5c5abe89e4ef1b4898d52eb40dc318f6f6a6193e74a94a8

4. Artifacts

- biougra.qasm: QASM 2.0 Bell circuit
- real_qpu.py: QPU execution script
- qubit_wallet.py: Deterministic wallet derivation from quantum measurement
- chain.json: Sealed chain
- human-qubit-machine.tar.gz: Full reproducible package (13KB)

5. Reproducibility

Requirements: Python 3, Qiskit, Linux/Termux environment

Command: python3 real_qpu.py

Expected: Bell distribution approx 50/50

6. License and Origin Tax

License: See LICENSE (1.4KB)

Origin Tax: 1% rule defined in ORIGIN.json

7. Conclusion

Protocol demonstrates low-resource quantum anchoring achievable on consumer mobile hardware with access to cloud QPUs.